

APM1110_FA7_DeRamos_AneezaAnn_and_Eduvala_Mikhael

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github link: https://github.com/AneezaAnn/APM1110_FA7_DeRamos_AneezaAnn_and_Eduvala_Mikhael.git

The chosen scenario is how long one student tries entering the gate when their Face ID won't work at FEU gate 3. Here are the gathered data:

1. 1 minute 23 seconds (83 seconds)
2. 0 minutes 54 seconds (54 seconds)
3. 0 minutes 51 seconds (51 seconds)
4. 1 minute 6 seconds (66 seconds)
5. 1 minute 21 seconds (81 seconds)
6. 0 minutes 42 seconds (42 seconds)
7. 0 minutes 24 seconds (24 seconds)
8. 0 minutes 38 seconds (38 seconds)
9. 0 minutes 26 seconds (26 seconds)
10. 0 minutes 31 seconds (31 seconds)
11. 0 minutes 52 seconds (52 seconds)
12. 0 minutes 17 seconds (17 seconds)
13. 0 minutes 48 seconds (48 seconds)
14. 0 minutes 22 seconds (22 seconds)
15. 0 minutes 10 seconds (10 seconds)
16. 1 minute 2 seconds (62 seconds)
17. 0 minutes 26 seconds (26 seconds)
18. 0 minutes 17 seconds (17 seconds)
19. 0 minutes 12 seconds (12 seconds)
20. 0 minutes 24 seconds (24 seconds)
21. 0 minutes 35 seconds (35 seconds)
22. 0 minutes 27 seconds (27 seconds)

23. 0 minutes 20 seconds (20 seconds)
24. 0 minutes 16 seconds (16 seconds)
25. 1 minute 14 seconds (74 seconds)
26. 0 minutes 33 seconds (33 seconds)
27. 0 minutes 21 seconds (21 seconds)
28. 0 minutes 25 seconds (25 seconds)
29. 0 minutes 18 seconds (18 seconds)
30. 0 minutes 14 seconds (14 seconds)

```
# Vector of observed times in seconds
```

```
faceid_delay_sec <- c(83, 54, 51, 66, 81, 42, 24, 38, 26, 31,
                     52, 17, 48, 22, 10, 62, 26, 17, 12, 24,
                     35, 27, 20, 16, 74, 33, 21, 25, 18, 14)
```

```
# Calculate descriptive statistics
```

```
avg_delay_sec <- mean(faceid_delay_sec)
mins_component <- avg_delay_sec %/% 60
secs_component <- avg_delay_sec %% 60
```

```
# Model Parameter Calculation
```

```
rate_lambda <- 1 / avg_delay_sec
```

```
# Evaluate cumulative probability for t <= 30 seconds
```

```
target_cutoff <- 30
prob_within_cutoff <- pexp(target_cutoff, rate = rate_lambda)
```

```
# Output summary to console
```

```
cat(sprintf("Average Failure Duration: %d minute(s) and %.2f second(s)", mins_component, secs_component,
```

```
## Average Failure Duration: 0 minute(s) and 35.63 second(s)
```

```
cat(sprintf("Rate Parameter (lambda): %.4f per second", rate_lambda))
```

```
## Rate Parameter (lambda): 0.0281 per second
```

```
cat(sprintf("Probability of resolving in <= 30 seconds: %.2f%%", prob_within_cutoff * 100))
```

```
## Probability of resolving in <= 30 seconds: 56.91%
```

Analysis & Practical Applications

While occasional biometric recognition issues seem minor in isolation, their cumulative impact during peak morning hours creates substantial pedestrian congestion. Our field observations indicate that when a scanner fails, a student spends up to 83 seconds attempting position adjustments or requesting manual clearance from security staff.

To mitigate these delays, administration should still implement a dual-authentication fallback—such (RFID tap card) and / or to take a new facial scan for those students who have trouble. These shall allow students to bypass stalled biometric turnstiles quickly.

Summary of Results

This investigation evaluated the operational impact of gate facial recognition errors across 30 sampled events. Standardizing all measurements into seconds yielded a sample mean (μ) of 35.63 seconds per failed attempt. Fitting an exponential distribution model resulted in an estimated rate parameter $\lambda = 0.0281 \text{ s}^{-1}$. Using the cumulative distribution function (CDF), we determined that approximately 57.00% of failed entry attempts are cleared or abandoned within 30 seconds.

```
# Re-assigning parameters for graphical plotting

rate_param <- 1 / mean(faceid_delay_sec)

# Constructing density histogram with theoretical overlay

hist(faceid_delay_sec,
     breaks = 9,
     freq = FALSE,
     col = "lightblue",
     border = "white",
     main = "Density Distribution of Face ID Entry Delays",
     xlab = "Duration of Failed Attempt (Seconds)",
     ylab = "Probability Density")

# Adding theoretical Exponential PDF Curve

curve(dexp(x, rate = rate_param),
     from = 0,
     to = max(faceid_delay_sec) + 10,
     col = "darkred",
     lwd = 2,
     add = TRUE)

# Legend positioning

legend("topright",
     legend = c("Observed Data", "Exponential Model"),
     fill = c("lightblue", NA),
     col = c(NA, "darkred"),
     lwd = c(NA, 2),
     bty = "n")
```

Density Distribution of Face ID Entry Delays

